Determining Continuity from Graphs

Introduction
Continuity at a Point
Continuity and Removable or Essential Discontinuities at a Given Point
Identifying Points of Discontinuity
Determining the Behavior of a Function in the Neighborhood of a Point

Prerequisites

Intervals Limits from Graphs

Introduction

A function $f(x)$ is continuous over its entire domain if the graph of $y = f(x)$ can be drawn using a pencil and a piece of paper without taking the pencil off of the paper.

To illustrate, let's consider the function $y = x^2$ whose graph is shown below.

We can draw as much of the function $y = x^2$ as we want without taking our pencil off of the paper. Therefore, we say that $y = x^2$ is **continuous** at every point in its domain.

Let's compare this to the function $y = \frac{1}{x}$ pictured below:

Notice that this function has a jump in its value at $x = 0$.

It is impossible to draw $y = \frac{1}{x}$ without taking the pencil off the paper to fill in the jump at $x = 0$. Therefore, we say that $y = \frac{1}{x}$ has a **discontinuity** at $x = 0$.

Continuity at a Point

Now consider the function $y = \lfloor x \rfloor$ below. This function has a discontinuity at $x = 0$, so it is not continuous over its entire domain.

However, the function $\lfloor x \rfloor$ may be continuous at other points, such as at $x = 1$.

To determine whether $\lfloor x \rfloor$ is continuous at $x = 1$, we have to consider points in the **neighborhood** of $x = 1$. Roughly speaking, the neighborhood of $x = 1$ means the points on the curve that lies at a coordinate that's close to 1.

Indeed, in the neighborhood of $x = 1$, we can draw the function $\lfloor x \rfloor$ without taking the pencil off the paper. So, $\lfloor x \rfloor$ is indeed continuous at $x = 1$.

Example: Determining Whether a Function Is Continuous at a Given Point

Is the function $y = f(x)$, plotted below, continuous at $x = 0$?

EXPLANATION

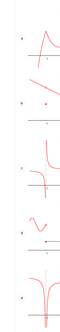
Because there is an asymptote at $x = 0$, it is impossible to draw the function in the neighborhood of $x = 0$ without taking our pencil off of the paper.

Therefore, the function $y = f(x)$ is not continuous at $x = 0$.

KEY PREREQUISITE
Piecewise Functions

Question 1

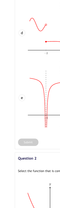
Select the function that is continuous at $x = -1$.



a

b

c



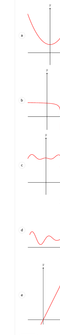
d

e

Submit

Question 2

Select the function that is continuous at $x = 1$.



a

b

c

d

e

Submit

Example: Identifying Points of Discontinuity

What are the points of discontinuity of the function $y = g(x)$, plotted below?

EXPLANATION

The points of discontinuity are at $x = 0$, $x = 1$, and at $x = 2.5$. It is impossible to draw the function in the neighborhood of these three points without taking our pencil off of the page.

Question 3

What are the points of discontinuity of the function plotted above?

a. $x = 0$ only

b. $x = 1$ only

c. $x = 0$ and $x = 1$

d. $x = 1$ and $x = 2$

e. $x = 1, 2, 3$

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Question 4

What are the points of discontinuity of the function plotted above?

a. $x = 0$ only

b. $x = 0$ and $x = 2$

c. $x = 0$ only

d. $x = 2$

e. $x = -2$ and $x = 0$

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Example: Determining the Behavior of a Function in the Neighborhood of a Point

The graph of $y = f(x)$ is shown above. Which of the following statements are true?

I. $f(x)$ is continuous at $x = 2$.

II. $\lim_{x \to 2} f(x)$ exists.

III. $f(x)$ is continuous at $x = 3$.

EXPLANATION

By inspecting the graph, we can conclude the following:

- Statement I is true. Indeed, $f(2) = 3$.
- Statement II is false. From the graph, we can see that the left and right sided limits are different:

$$\lim_{x \to 2^-} f(x) = 1,$$
$$\lim_{x \to 2^+} f(x) = 3.$$

Therefore, $\lim_{x \to 2} f(x)$ does not exist.

- Statement III is false. The graph of the function breaks at $x = 3$. It's impossible to draw the graph of $f(x)$ without taking our pencil off of the paper. Therefore, $f(x)$ is not continuous at $x = 3$.

In conclusion, only statement I is true.

KEY PREREQUISITE(S)
Finding the Existence of a Limit Using One Sided Limits
Infinite Limits from Graphs

Question 5

Which of the following statements are true with respect to the function $y = f(x)$ whose graph is plotted above.

I. $\lim_{x \to 0} f(x)$ is finite

II. $\lim_{x \to 0} f(x)$ is finite

III. $f(x)$ has a discontinuity at $x = 2$

a. II and III only

b. I only

c. II only

d. I and II only

e. III only

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Question 6

The graph of $y = f(x)$ is shown above. Which of the following statements are true?

I. $f(x)$ is defined at $x = 3$

II. $\lim_{x \to 3} f(x)$ exists

III. $f(x)$ is continuous at $x = 3$

a. I only

b. I and II only

c. I and III only

d. III only

e. II and III only

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